

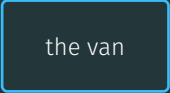
Reinforcement Learning

from one delivery van to the Bellman equation

Carlos Cotrini, ETH Zurich

Motivation

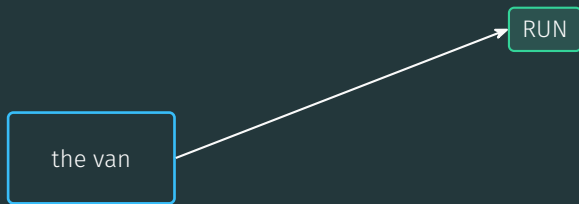
One van, one call a week



the van

every week: exactly one call

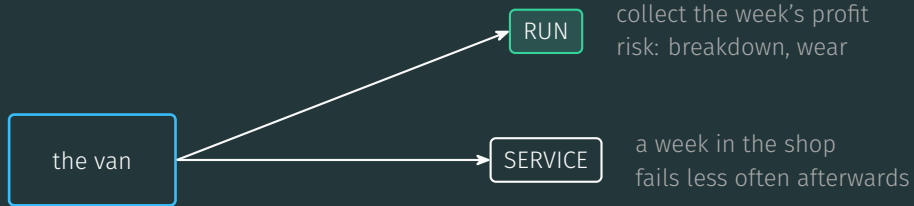
One van, one call a week



collect the week's profit
risk: breakdown, wear

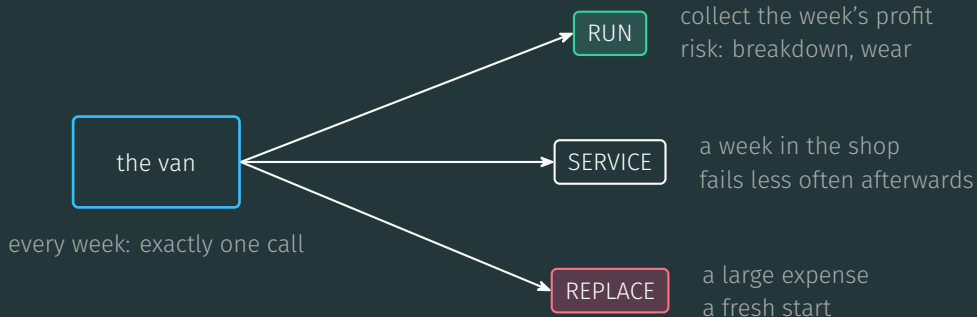
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One van, one call a week

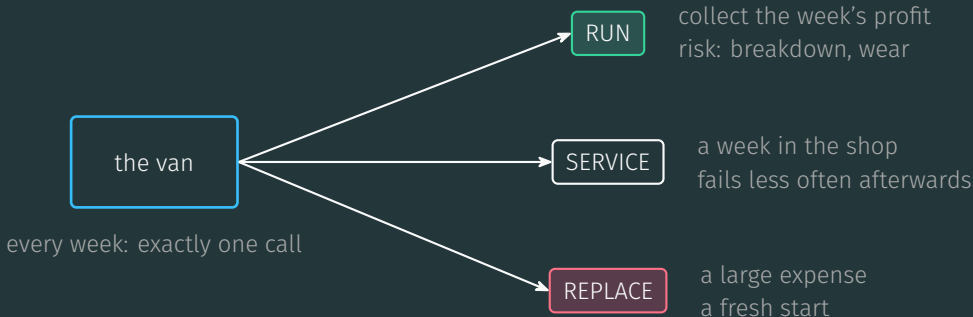


every week: exactly one call

One van, one call a week



One van, one call a week



short-term cost against long-term payoff, chance at every step

Markov decision processes

The model: four ingredients

S: states

four wear
levels

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A: actions

three
calls

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T: dynamics

how the world
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$\gamma = 0.9$

how much the
future counts

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$$(s', r) = T(s, a)$$

The model: four ingredients

S: states

four wear
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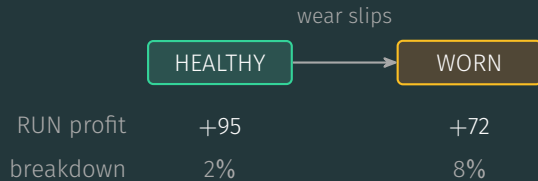
$$\left(\underbrace{s'}_{\text{the state next week}}, \underbrace{r}_{\text{the reward}} \right) = T(s, a)$$

The fleet sheet

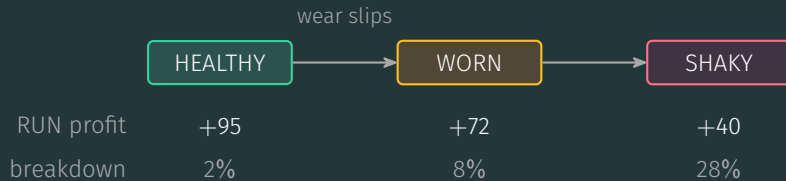
HEALTHY

RUN profit	+95
breakdown	2%

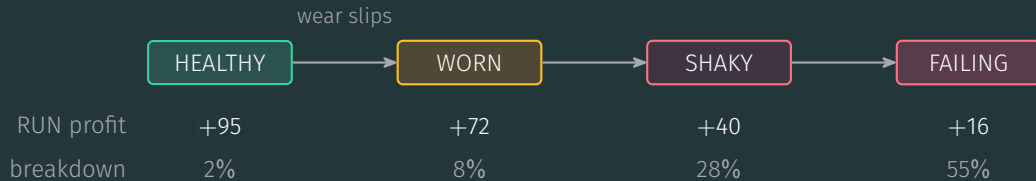
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The fleet sheet



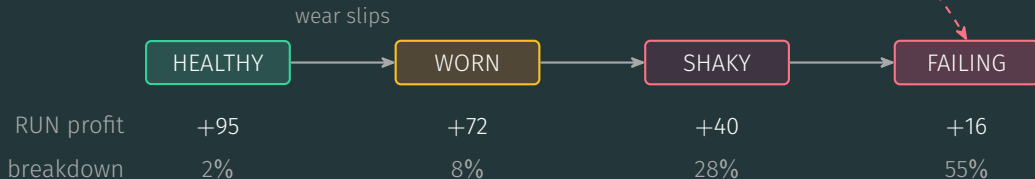
The fleet sheet



The fleet sheet

breakdown under RUN: one net reward, profit minus 280

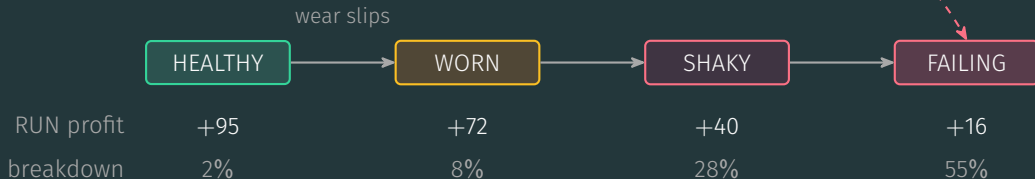
from HEALTHY: $95 - 280 = -185$



The fleet sheet

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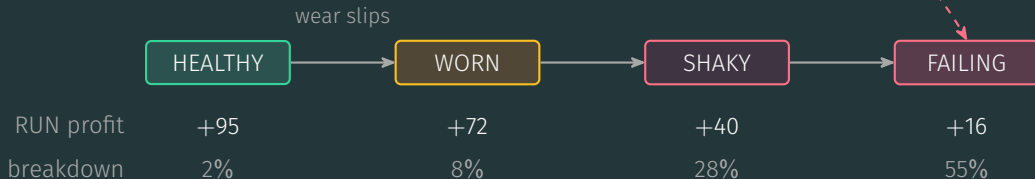


SERVICE: -50, a week in the shop (WORN back to HEALTHY 95% of the time)

The fleet sheet

breakdown under RUN: one net reward, profit minus 280

from HEALTHY: $95 - 280 = -185$



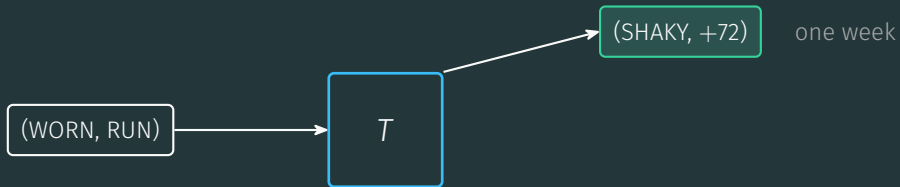
SERVICE: -50, a week in the shop (WORN back to HEALTHY 95% of the time)

REPLACE: -130, a brand-new HEALTHY van next week

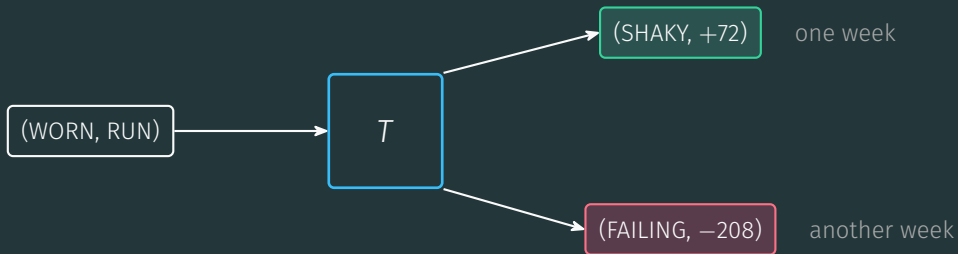
Same pair in, different week out



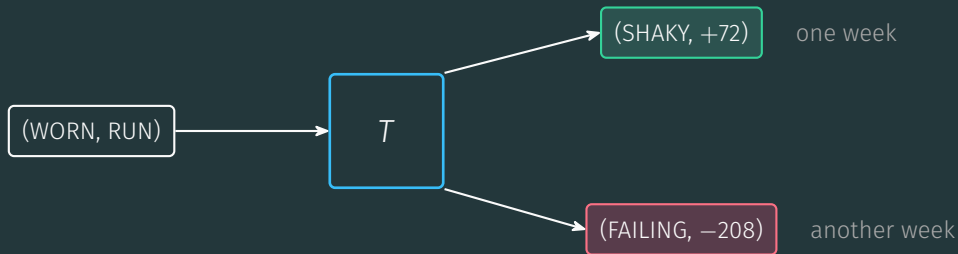
Same pair in, different week out



Same pair in, different week out



Same pair in, different week out



same input, different output: the dynamics T is random

Policies and trajectories

Playbook in, tape out

a policy (the playbook): $\pi : S \rightarrow A$, one call per state

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HEALTHY $\rightarrow$ RUN

WORN $\rightarrow$ SERVICE

SHAKY $\rightarrow$ REPLACE

FAILING $\rightarrow$ REPLACE

each week: $a_i = \pi(s_i)$, then $(s_{i+1}, r_i) = T(s_i, a_i)$

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wk 1
HEALTHY, RUN
+95

Playbook in, tape out

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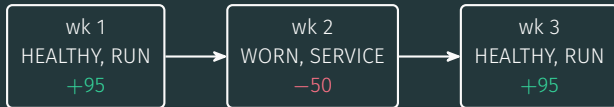
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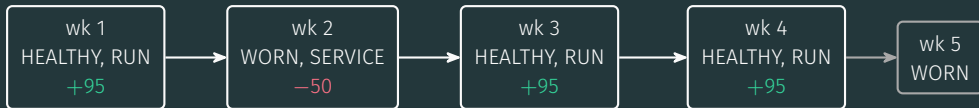
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each week: $a_i = \pi(s_i)$, then $(s_{i+1}, r_i) = T(s_i, a_i)$



the trajectory: $s_1, a_1, r_1, s_2, a_2, r_2, \dots$ one week per step, forever

Playbook in, tape out

a policy (the playbook): $\pi : S \rightarrow A$, one call per state

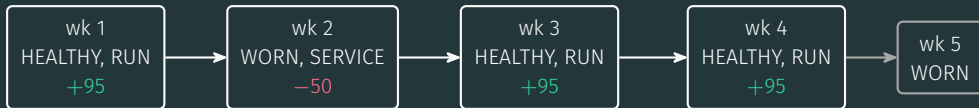
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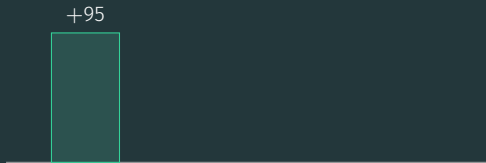
the trajectory: $s_1, a_1, r_1, s_2, a_2, r_2, \dots$ one week per step, forever

the tape is random; another run, same start: (HEALTHY, RUN, -185) $\rightarrow$ FAILING

The goal: accumulating reward

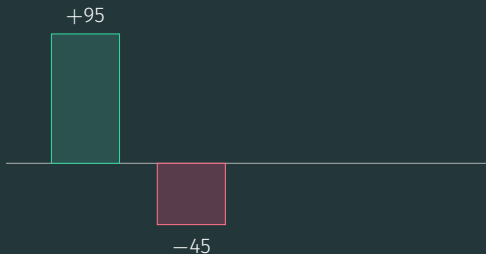
The return: one number per tape

$$G_1 = r_1$$



The return: one number per tape

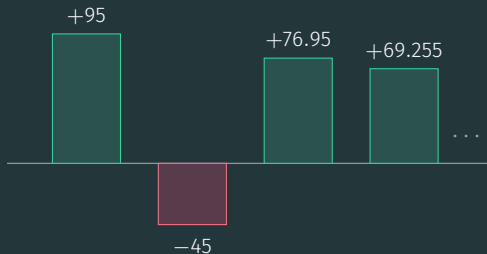
$$G_1 = \underbrace{r_1}_{\text{in full}} + \underbrace{\gamma r_2}_{\text{at 90\%}}$$



each week shrunk
by $\gamma = 0.9$

The return: one number per tape

$$G_1 = \underbrace{r_1}_{\text{in full}} + \underbrace{\gamma r_2}_{\text{at 90\%}} + \underbrace{\gamma^2 r_3}_{\text{at 81\%}} + \dots$$

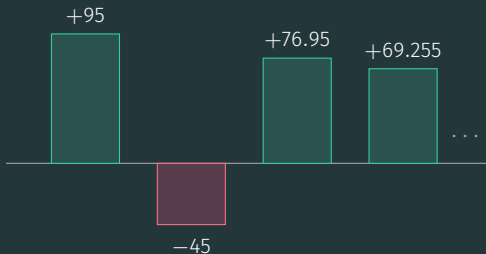


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$$G_1 = 95 + 0.9(-50) + 0.81 \cdot 95 + 0.729 \cdot 95 + \dots = 95 - 45 + 76.95 + 69.255 + \dots$$

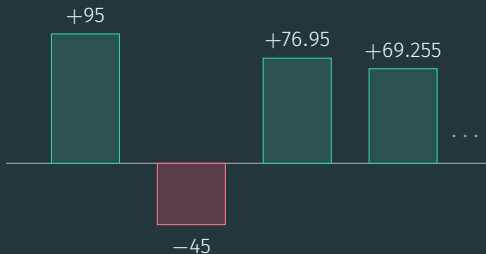


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each week shrunk
by $\gamma = 0.9$

the shrinking factors keep
the never-ending sum finite

The recurrence

$$G_i = r_i + \gamma r_{i+1} + \gamma^2 r_{i+2} + \dots$$

r_i

r_{i+1}

r_{i+2}

$\dots$

The recurrence

$$G_i = r_i + \gamma(r_{i+1} + \gamma r_{i+2} + \dots)$$



G_{i+1} , discounted once by γ

The recurrence

$$G_i = \underbrace{r_i}_{\text{this week}} + \gamma \underbrace{G_{i+1}}_{\text{everything after}}$$



G_{i+1} , discounted once by γ

The recurrence

$$G_i = \underbrace{r_i}_{\text{this week}} + \gamma \underbrace{G_{i+1}}_{\text{everything after}}$$



G_{i+1} , discounted once by γ

this week's reward $+ \gamma \times$ all the rest

Maximize the average

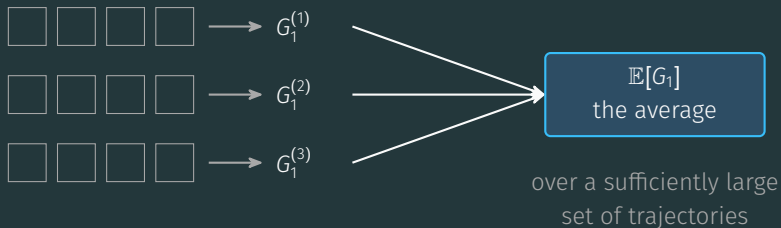
 $\longrightarrow G_1^{(1)}$

 $\longrightarrow G_1^{(2)}$

 $\longrightarrow G_1^{(3)}$

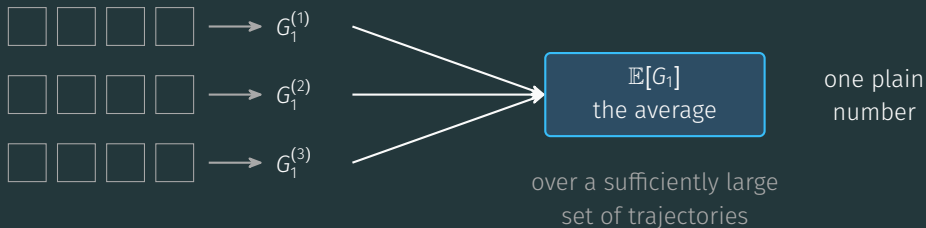
same playbook, same start: a different G_1 every run

Maximize the average



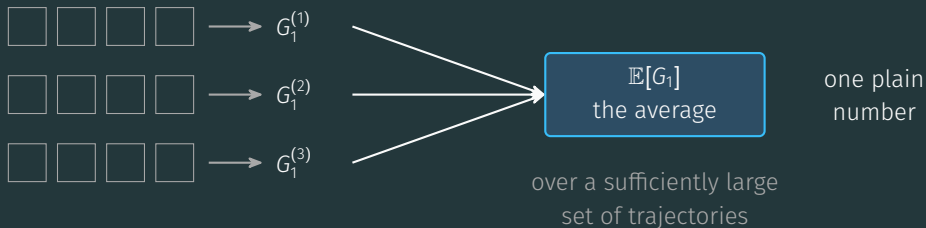
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Maximize the average



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Maximize the average



same playbook, same start: a different G_1 every run

the goal: find the policy π that maximizes $\mathbb{E}[G_1]$

The Q -function and the Bellman equation

The scorecard Q^*

$$Q^*(s, a) = \mathbb{E}[G_i \mid s_i = s, a_i = a]$$

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$$Q^*(s, a) = \mathbb{E} \left[G_i \mid \underbrace{s_i = s, a_i = a}_{\text{in } s, \text{ play } a, \text{ optimal afterwards}} \right]$$

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$$\pi^*(s) = \arg \max_a Q^*(s, a) \quad \text{read each row, keep the best}$$

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$\pi^*(s) = \arg \max_a Q^*(s, a)$ read each row, keep the best

	RUN	SERVICE	REPLACE
HEALTHY	311.0	229.9	149.9

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	RUN	SERVICE	REPLACE
HEALTHY	311.0	229.9	149.9
WORN	203.4	226.1	149.9

The scorecard Q^*

$$Q^*(s, a) = \mathbb{E}[G_i \mid \underbrace{s_i = s, a_i = a}_{\text{in } s, \text{ play } a, \text{ optimal afterwards}}]$$

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	RUN	SERVICE	REPLACE
HEALTHY	311.0	229.9	149.9
WORN	203.4	226.1	149.9
SHAKY	96.5	126.4	149.9

The scorecard Q^*

$$Q^*(s, a) = \mathbb{E}[G_i \mid \underbrace{s_i = s, a_i = a}]$$

in s , play a , optimal afterwards

$\pi^*(s) = \arg \max_a Q^*(s, a)$ read each row, keep the best

	RUN	SERVICE	REPLACE
HEALTHY	311.0	229.9	149.9
WORN	203.4	226.1	149.9
SHAKY	96.5	126.4	149.9
FAILING	-3.1	86.9	149.9

The scorecard Q^*

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in s , play a , optimal afterwards

$\pi^*(s) = \arg \max_a Q^*(s, a)$ read each row, keep the best

	RUN	SERVICE	REPLACE
HEALTHY	311.0	229.9	149.9
WORN	203.4	226.1	149.9
SHAKY	96.5	126.4	149.9
FAILING	-3.1	86.9	149.9

149.9 beats 126.4 by 23.5

the optimal playbook replaces a van that still starts every morning

The Bellman equation

$$G_i = r_i + \gamma G_{i+1} \quad (\text{every single week})$$



The Bellman equation

$$Q^*(s_i, a_i) = \mathbb{E}[r_i + \gamma G_{i+1}] \quad (\text{average both sides})$$



The Bellman equation

$$Q^*(s_i, a_i) = \mathbb{E} \left[\underbrace{r_i}_{\text{this week's cash}} + \gamma \underbrace{Q^*(s_{i+1}, a_{i+1})}_{\text{next week's value}} \right]$$

replacing G_{i+1} by its average $Q^*(s_{i+1}, a_{i+1})$: averaging in two rounds



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replacing G_{i+1} by its average $Q^*(s_{i+1}, a_{i+1})$: averaging in two rounds



one equation per cell: twelve equations pin down the table

SARSA: learning from the logbook

No odds? Average the logbook

the Bellman average needs the odds, and nobody prints them for your van

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(WORN, RUN) paid:

+72

+72

+72

-208

+72

...

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$$q(s_i, a_i) \leftarrow q(s_i, a_i) + \alpha [r_i - q(s_i, a_i)]$$

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the Bellman average needs the odds, and nobody prints them for your van

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+72

...

$$q(s_i, a_i) \leftarrow \underbrace{q(s_i, a_i)}_{\text{old guess}} + \underbrace{\alpha}_{0.15} \left[\underbrace{r_i}_{\text{what the week paid}} - q(s_i, a_i) \right]$$

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$$q(s_i, a_i) \leftarrow \underbrace{q(s_i, a_i)}_{\text{old guess}} + \underbrace{\alpha}_{0.15} \underbrace{[r_i - q(s_i, a_i)]}_{\text{the miss: move a fraction toward it}}$$

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exponential smoothing: new forecast = old forecast + $\alpha \times (\text{actual} - \text{old forecast})$

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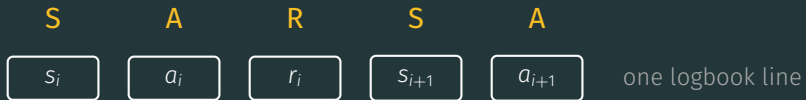
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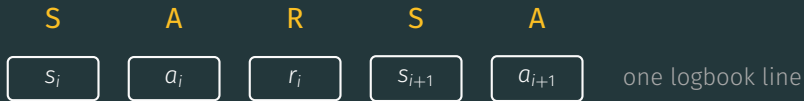
exponential smoothing: new forecast = old forecast + $\alpha \times (\text{actual} - \text{old forecast})$

nudge, never overwrite: one bad week must not erase a year

The actual grows up: SARSA

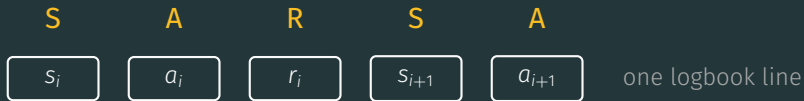


The actual grows up: SARSA



$$q(s_i, a_i) \leftarrow q(s_i, a_i) + \alpha [r_i - q(s_i, a_i)] \quad (\text{cash only})$$

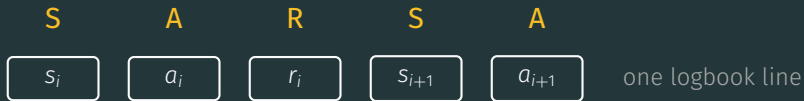
The actual grows up: SARSA



$$q(s_i, a_i) \leftarrow q(s_i, a_i) + \alpha \left[\underbrace{r_i + \gamma q(s_{i+1}, a_{i+1})}_{\text{the actual, grown up}} - q(s_i, a_i) \right]$$

the future has not happened yet: the guess $q(s_{i+1}, a_{i+1})$ stands in for Q^*

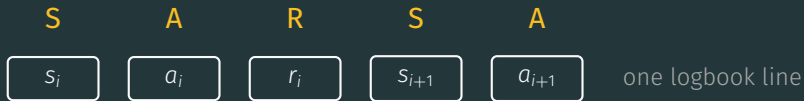
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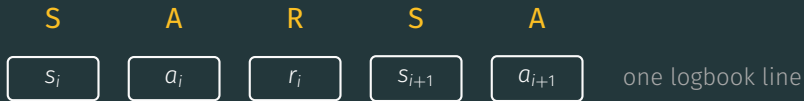


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the future has not happened yet: the guess $q(s_{i+1}, a_{i+1})$ stands in for Q^*

$\alpha = 0.15$, constant · ϵ -greedy: a random call in 15% of weeks · a_{i+1} : actually executed, never a max

The actual grows up: SARSA



$$q(s_i, a_i) \leftarrow q(s_i, a_i) + \alpha \underbrace{\left[r_i + \gamma q(s_{i+1}, a_{i+1}) - q(s_i, a_i) \right]}_{\text{the surprise}}$$

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$\alpha = 0.15$, constant · ϵ -greedy: a random call in 15% of weeks · a_{i+1} : actually executed, never a max

Bellman, with the expectation replaced by an arriving average

One week, one nudge

		RUN	SERVICE	REPLACE
twelve guesses, all start at 0	HEALTHY	0	0	0
	WORN	0	0	0
	SHAKY	0	0	0
	FAILING	0	0	0

One week, one nudge

week one in the logbook: (HEALTHY, RUN, +95, WORN, SERVICE)

twelve guesses, all start at 0		RUN	SERVICE	REPLACE
	HEALTHY	0	0	0
	WORN	0	0	0
	SHAKY	0	0	0
	FAILING	0	0	0

One week, one nudge

week one in the logbook: (HEALTHY, RUN, +95, WORN, SERVICE)

$$\text{evidence} = r_i + \gamma q(\text{WORN}, \text{SERVICE}) = 95 + 0.9 \times 0 = 95$$

		RUN	SERVICE	REPLACE
twelve guesses, all start at 0	HEALTHY	0	0	0
	WORN	0	0	0
	SHAKY	0	0	0
	FAILING	0	0	0

One week, one nudge

week one in the logbook: (HEALTHY, RUN, +95, WORN, SERVICE)

$$\text{evidence} = r_i + \gamma q(\text{WORN}, \text{SERVICE}) = 95 + 0.9 \times 0 = 95$$

$$\text{surprise} = 95 - 0 = 95$$

nudge: $+0.15 \times 95$



twelve guesses, all start at 0

		RUN	SERVICE	REPLACE
HEALTHY		0	0	0
WORN		0	0	0
SHAKY		0	0	0
FAILING		0	0	0

One week, one nudge

week one in the logbook: (HEALTHY, RUN, +95, WORN, SERVICE)

$$\text{evidence} = r_i + \gamma q(\text{WORN}, \text{SERVICE}) = 95 + 0.9 \times 0 = 95$$

$$\text{surprise} = 95 - 0 = 95$$

nudge: $+0.15 \times 95$



twelve guesses, all start at 0		RUN	SERVICE	REPLACE
one cell moves, eleven stay put	HEALTHY	14.25	0	0
	WORN	0	0	0
	SHAKY	0	0	0
	FAILING	0	0	0

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the cells settle below Q^* and keep jittering, but the argmax per row comes out right

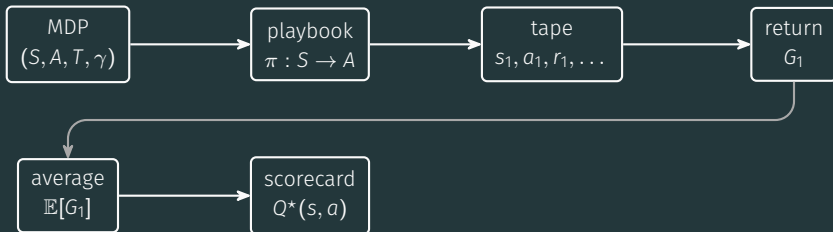
The whole story



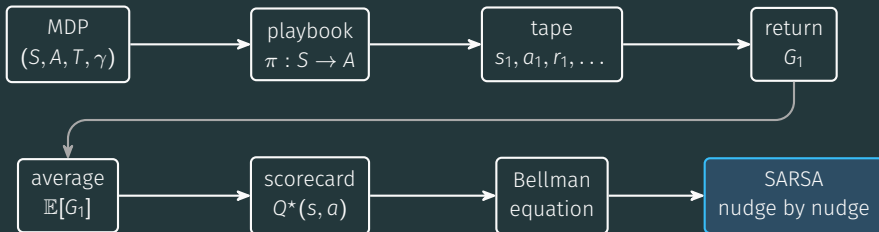
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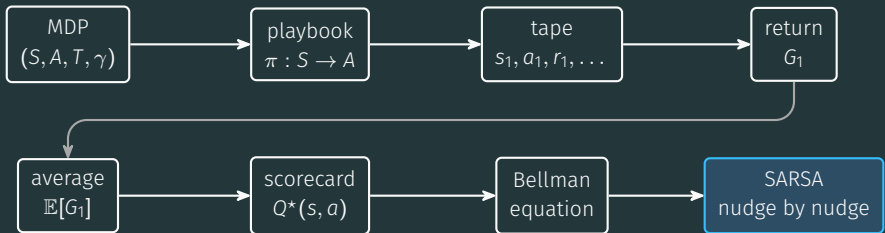
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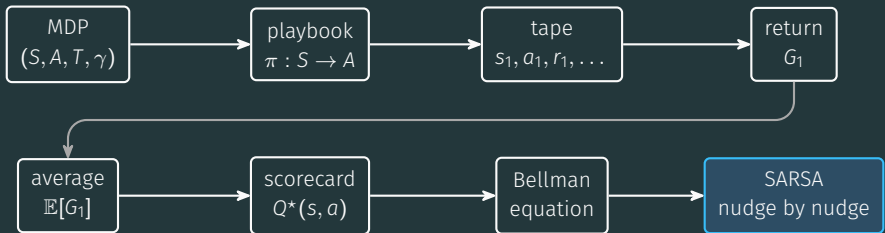


The whole story



SARSA solves the Bellman equation by averaging evidence instead of computing expectations

The whole story



SARSA solves the Bellman equation by averaging evidence instead of computing expectations

play it: https://sml-fs26.github.io/classic_rl/repair-or-replace/

Appendix: why the nudge works

The Robbins-Monro iteration

find the number x^* that equals the average of noisy samples, one at a time

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SARSA is Robbins-Monro on the Bellman equation, cell by cell

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A_n : the average of the first n observations $x_1, \dots, x_n$

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SARSA is a running average of the weekly evidence